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CKS Exam Cram Sheet
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RBAC

```
kubectl create role pod-reader --verb=get,list,watch --resource=pods -n dev --dry-run=client -o yaml > role.yaml
kubectl create rolebinding read-pods --role=pod-reader --user=alice -n dev --dry-run=client -o yaml > rb.yaml
kubectl create clusterrole secret-admin --verb=get,list,watch --resource=secrets --dry-run=client -o yaml > cr.yaml
kubectl create clusterrolebinding secret-admin-binding --clusterrole=secret-admin --serviceaccount=dev:sa1 -o yaml > crb.yaml
```

Service Account & Pod

```
apiVersion: v1
kind: ServiceAccount
metadata: { name: sa1, namespace: dev }
```

spec:

```
  serviceAccountName: sa1
  containers: [{ name: nginx, image: nginx }]
```

Security Context

```
securityContext:
  runAsNonRoot: true
  allowPrivilegeEscalation: false
  capabilities: { drop: ["ALL"], add: ["NET_BIND_SERVICE"] }
```

Network Policy (default deny)

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
spec:
  podSelector: {}
  policyTypes: [Ingress, Egress]
```

Secrets

```
kubectl create secret generic db-secret --from-literal=username=admin --from-literal=password=pass123 -n dev
```

Ingress with TLS

```
kubectl create secret tls tls-secret --cert=cert.crt --key=cert.key -n dev
```

Audit Policy

```
--audit-policy-file=/etc/kubernetes/audit-policy.yaml
--audit-log-path=/var/log/apiserver/audit.log
```

Encryption at Rest

```
--encryption-provider-config=/etc/kubernetes/encryption-config.yaml
```

etcdctl

```
etcdctl --endpoints=https://127.0.0.1:2379 --cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
etcdctl snapshot save /var/lib/etcd/snapshot.db
etcdctl snapshot restore /var/lib/etcd/snapshot.db --data-dir=/var/lib/etcd
```

kube-apiserver Flags

```
--authorization-mode=Node, RBAC
--encryption-provider-config=/etc/kubernetes/encryption-config.yaml
--tls-cert-file=/etc/kubernetes/pki/apiserver.crt
--tls-private-key-file=/etc/kubernetes/pki/apiserver.key
--client-ca-file=/etc/kubernetes/pki/ca.crt
```

kubelet Flags

```
--anonymous-auth=false
--authorization-mode=Webhook
--authentication-token-webhook=true
```

Pod Security Admission

```
kubectl label ns dev pod-security.kubernetes.io/enforce=restricted
```

Admission Plugins

--enable-admission-plugins=AlwaysPullImages,PodSecurity,NodeRestriction

Quick Helpers

kubectl run tmp --rm -it --image=busybox -- /bin/sh

kubectl auth can-i get pods --as=system:serviceaccount:dev:sa1 -n dev

kubectl describe pod <pod>

kubectl logs <pod>

kubectl exec -it <pod> -- sh

File Paths

/etc/kubernetes/pki/ca.crt

/etc/kubernetes/pki/apiserver.crt

/etc/kubernetes/pki/apiserver.key

/etc/kubernetes/pki/etcd/ca.crt

/etc/kubernetes/pki/etcd/server.crt

/etc/kubernetes/pki/etcd/server.key

/etc/kubernetes/encryption-config.yaml

/etc/kubernetes/audit-policy.yaml

/etc/kubernetes/admin.conf